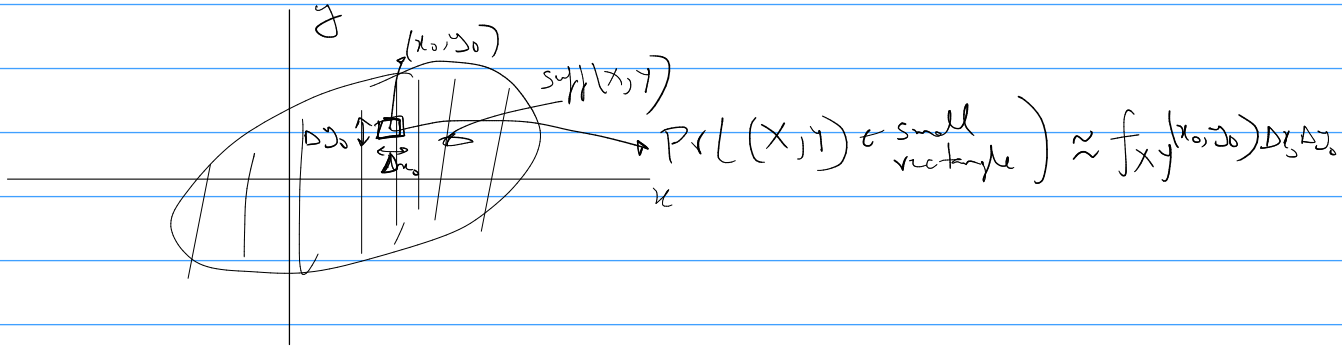
Slp 8 $X, Y \sim$ PDF $f_{XY}(x,y)$ $X, Y$ : jointly Continuous with PDF $f_{XY}$

$$\text{Supp}(X,Y) = \{(x,y) : f_{XY}(x,y) > 0\}$$

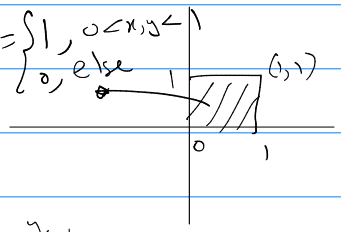



Ex: $X, Y \sim$ Uniform$(S)$ $S \subseteq \mathbb{R}^2$

?

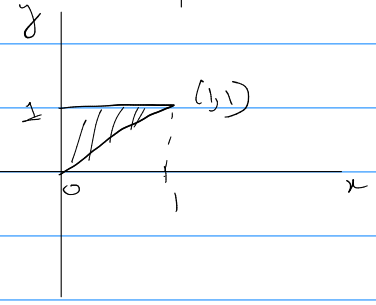
- $S = \{(x,y) : 0 < x,y < 1\}$

$$f_{XY}(x,y) = \begin{cases} 1, & 0<x,y<1 \\ 0, & \text{else} \end{cases}$$




- $S = \{(x,y) : 0 < x < y < 1\}$

$$f_{XY}(x,y) = \begin{cases} 2, & 0<x<y<1 \\ 0, & \text{else} \end{cases}$$

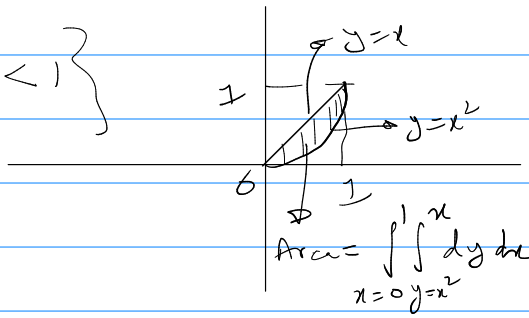



- $S = \{(x,y) : 0 < x^2 < y < x < 1\}$

$$\Updownarrow$$

$$0 < x < 1, \quad x^2 < y < x$$

$$f_{XY}(x,y) = \begin{cases} 6, & 0<x<1, \ x^2<y<x, \\ 0, & \text{else} \end{cases}$$

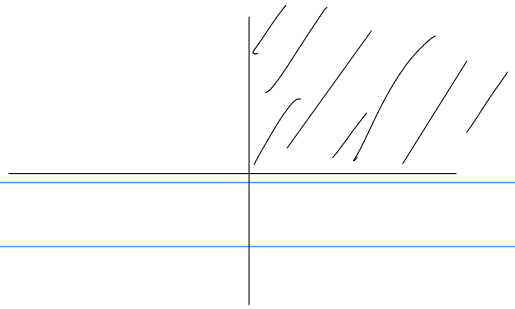



$$\text{Area} = \int_{x=0}^{1} \int_{y=x^2}^{x} dy\, dx$$

$$= \int_0^1 (x - x^2)\, dx$$

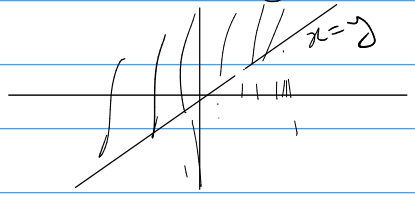
$$= 1/2 - 1/3 = 1/6$$

Ex: $f_{XY}(x,y) = \begin{cases} e^{x-y}, & x,y>0 \\ 0, & \text{else} \end{cases}$



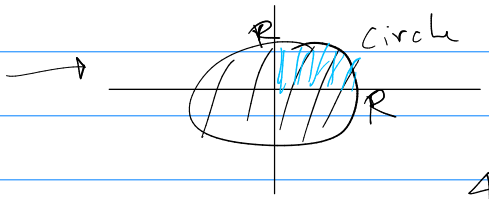
Probability of region: $X,Y \sim f_{XY}$

$$\rightarrow P(X<Y) = \int_{x=-\infty}^{\infty}\int_{y=x}^{\infty} f_{XY}(x,y)\,dy\,dx$$




$$= \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{y} f_{XY}(x,y)\,dx\,dy$$

$\rightarrow$ circle




$f_{XY} \propto e^{-\frac{(x^2+y^2)}{2}}$ all $x,y$

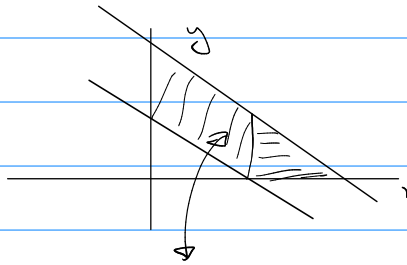
$$4\int_{x=0}^{r}\int_{y=0}^{\sqrt{r^2-x^2}} \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}}\,dy\,dx = 4\int_{r=0}^{R}\int_{\theta=0}^{\frac{\pi}{2}} \frac{1}{2\pi} e^{-\frac{r^2}{2}} r\,dr\,d\theta$$

$$= \int_0^R r e^{-r^2/2}\,dr$$

$$= 1 - e^{-R^2/2}$$

$\rightarrow 1 < X+Y < 2$ $\qquad f_{XY} = e^{-x-y}$, $x,y>0$

$e^{-1} - e^{-2}$



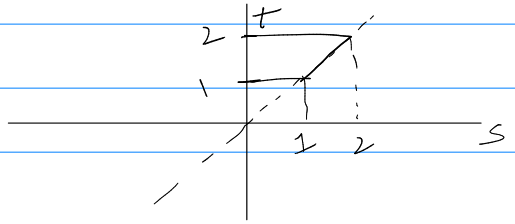

$t = x+y$

$s = x$

$|J| = 1$

$x>0,\ y>0,\ x+y>1,\ x+y<2$

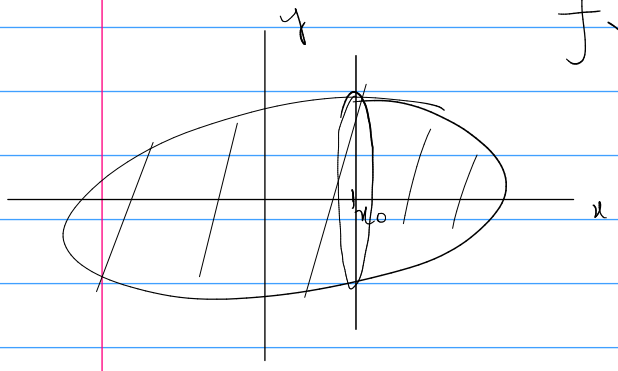
$s>0,\ t>s,\ t>1,\ t<2$



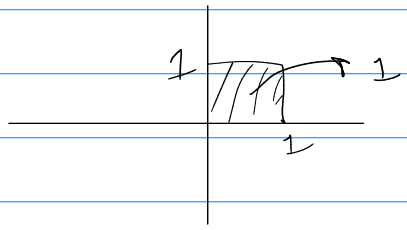
## Marginal PDFs

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{XY}(x,y)\,dy$$

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{XY}(x,y)\,dx$$

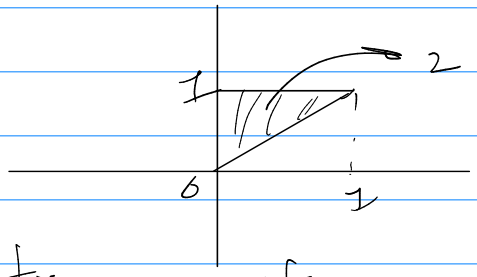



Ex:

① 

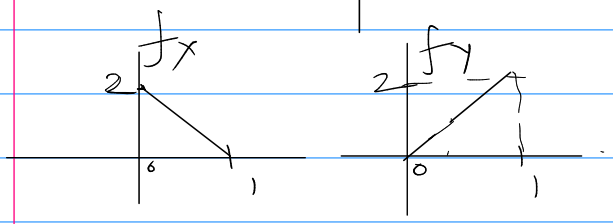

$X \sim \text{Uniform}[0,1]$

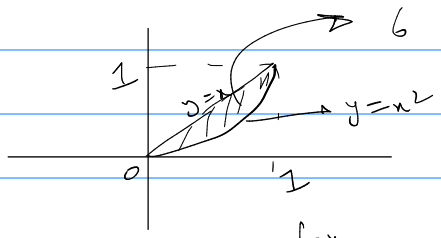
$Y \sim \text{Uniform}[0,1]$

② 


$X \sim \text{PDF} \begin{cases} 2(1-x) & 0<x<1 \\ 0, & \text{else} \end{cases}$

$Y \sim \text{PDF} \begin{cases} 2y & 0<y<1 \\ 0, & \text{else} \end{cases}$




③ 


$X \sim \text{PDF} \begin{cases} 6x(1-x) & 0<x<1 \\ 0, & \text{else} \end{cases}$

$Y \sim \text{PDF} \begin{cases} 6(\sqrt{y}-y), & 0<y<1 \\ 0, & \text{else} \end{cases}$

